

Math 350 Section 1
Fall 2024
Final Exam
December 16, 2024
Time Limit: 1 Hour 50 Minutes

Name (Print): _____

Student ID: _____

This exam contains 8 pages (including this cover page) and 7 problems. Check to see if any pages are missing. Enter all requested information on the top of this page, and put your initials on the top of every page, in case the pages become separated.

You may *not* use your books or notes on this exam.

You are required to show your work on each problem on this exam. The following rules apply:

- **Organize your work**, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little credit.
- **Mysterious or unsupported answers will not receive full credit.** A correct answer, unsupported by calculations, explanation, or algebraic work will receive no credit; an incorrect answer supported by substantially correct calculations and explanations might still receive partial credit. This especially applies to limit calculations.
- If you need more space, use the back of the pages; clearly indicate when you have done this.
- **Box Your Answer** where appropriate, in order to clearly indicate what you consider the answer to the question to be.
- **Default metric:** unless otherwise stated, the set $\mathbb{R}$ or $\mathbb{R}^n$ will be considered a metric space with the Euclidean metric

Problem	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
7	10	
Total:	70	

Do not write in the table to the right.

1. (10 points) **TRUE or FALSE!** Write TRUE if the statement is true. Otherwise, write FALSE. Your response should be in ALL CAPS. No justification is required.

(a) Every subset of $\mathbb{R}$ with the discrete metric is open

(b) The function $f(x) = e^x + x$ has an x -intercept

(c) If $f(x)$ is differentiable at $x = 0$, then so is $f(x^2)$

(d) The set of all irrational numbers is uncountable

(e) In any metric space, compact sets must be closed

2. (10 points) Let (M, d) be a metric space and $K \subseteq M$.

(a) Write the definition of K being compact

(b) Write the definition of K being bounded

(c) Prove that if K is not bounded, then it cannot be compact

3. (10 points) In this problem, let $\alpha(x) = x$ and

$$f(x) = \begin{cases} 1, & x \neq 3 \\ 5, & x = 3 \end{cases}$$

Carefully prove f is integrable with respect to α on $[0, 3]$ and determine the value of $\int_0^3 f d\alpha$

4. (10 points)

Consider the function

$$f(x) = \begin{cases} x, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q} \end{cases}$$

(a) Prove that $f(x)$ is not continuous at $x = c$ for $c \neq 0$

(b) Prove that $f(x)$ is continuous at $x = 0$

5. (10 points)

(a) Carefully explain why there is no continuous, surjective function $f : [0, 1] \rightarrow (0, 1]$

(b) Carefully explain why there is no continuous, surjective function $f : [0, 1] \rightarrow [0, 1/2) \cup (1/2, 1]$

(c) Show that there is a continuous, surjective function $f : (0, 1] \rightarrow [0, 1]$

6. (10 points) Let $\{x_n\}$ be a monotone increasing sequence of real numbers.

(a) State the Completeness Axiom

(b) Prove that if $\{x_n\}$ is also bounded above, then it converges.

7. (10 points) Suppose that $f(x)$ is a polynomial with roots at $x = 1$, $x = 2$, and $x = 3$

(a) State the Mean Value Theorem

(b) Prove that $f'(x)$ has at least two different real roots

(c) Prove that $f''(x)$ has at least one real root